PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

KILN HEAD POST 窑头岗位

MTI-CHP-CPM-SOP-004_Rev. 000

Information Security Classification
B – Business Information

Rev.	Revision Status	Date	Prepared by	Reviewed by	Approved by
0	Issued for Use	24-APR-2024	*Approval as attached*	*Approval as attached*	*Approval as attached*
C0	Issued for Approval	24-APR-2024	Song Yuhua	Li Jianjun	Clark Wu
B0	Issued for Review	24-APR-2024	Song Yuhua	Li Jianjun	
A0	Issued for Internal Discipline Review	24-APR-2024	Song Yuhua		

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

DOCUMENT CHANGE STATUS

No.	Page	Section	Change Log	Revision by	Revision date	Revision version
1.						
2.						
3.						

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

Table of Contents

DOCUMENT CHANGE STATUS 2

Table of Contents 3

1. Purpose 目的 4
2. Functional Description 功能说明 4
3. References 参考 4
4. PPE 个人保护装备 4
5. Hazards & Control 危害与控制 5
6. Equipment 工具/ 设备 6
7. Procedure 程序 6
8. Kiln Head Post 窑头岗位 Questionnaire 18
9. Appendix 19

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the kiln head post under complete closure.

此程序涵盖操作员对完全关闭下的窑头岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the kiln head post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the kiln head post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对窑头岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定窑头岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP岗位操作规程
- Rolary Kiln(4303-KLN-001). 回转窑(4304-KLN-001)
- Kiln Head Cooling Fan(4304-BLO-005/006). 窑头冷却风机(4304-BLO-005/006)
- Kiln Head Cover(4304-KHC-001). 窑头罩(4304-KHC-001)
- Gas Bumer(4304-BUR-001). 煤气燃烧器(4304-BUR-001)
- Combustion Fan(4304-BLO-001/002). 燃烧风机(4304-BLO-001/002)

4. PPE 个人保护装备



耳塞　安全帽　眼镜　劳保鞋　手套　工作服



防毒口罩

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue危害	Control Strategy 控制
1. Mechanical injury 2. Dust 3. Electric shock 4. Slippery conditions and a trip to danger. 5. Asphyxia 6. Empyrosis 7. Object falling from high altitudes. 8. Poisoning 9. Noise 10. High temperature. 1. 机械伤害； 2. 灰尘； 3. 触电； 4. 湿滑的条件，绊倒危险； 5. 窒息； 6. 烫伤； 7. 高空坠物； 8. 中毒； 9. 噪音； 10. 高温。	1. Install a protective cover on the rotating part of the equipment. 2. Wear a dust mask as required. 3. Do not contact the electrical switch with wet hands during operation, Regular inspection of the equipment leakage protection device is complete, effective. 4. Please hold the handrail up and down the stairs. 5. Clear clearance or maintenance operations. 6. Wear good labor protection goods. 7. Wear labor protection goods and check the safety of stacking items on the operation platform. 8. Leave the high temperature environment and rest in a cool and ventilated place. 9. Wear antigas masks as required. 10. Wear earplugs to protect your hearing. 1. 设备的转动部位安装防护罩； 2. 按要求佩戴防尘口罩； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 清仓或维修作业执行审批程序； 6. 穿戴好劳保护品； 7. 穿戴好劳保护品，检查各平台的物品堆放安全性； 8. 脱离高温环境，在阴凉通风处休息； 9. 按要求佩戴防毒口罩； 10. 佩戴好耳塞，保护听力。

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

6. Equipment 工具/ 设备

4304-LUS-001	Lubrication System润滑系统
	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训; 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患; 3. 检查并打扫设备、现场卫生; 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Rolary Kiln (4303-KLN-001) 检查回转窑(4304-KLN-001)	1. Check whether the bolts are tightened; 2. Check whether the rotary kiln lubrication system (4304-LUS-001) is in good condition, and whether the grease of each lubrication point is fully added; 3. Check whether the cooling water of the rotary kiln lubrication system (4304-LUS-001) is normal. 4. Check the kiln head seal is not allowed to have a stuck place; 5. Check whether the wheel adjustment is correct, whether the wheel and the surface of the ring belt uniform contact; 6. The system equipment and instrument power transmission are ready, and the central control confirms that the DCS system is normal; 7. Check the thermocouple is normal, its protective sleeve highlights the kiln lining surface, the thermocouple and the casing along the flush; 8. Check whether the pressure instrument device in the rotary kiln is normal and whether the signal transmission is normal; 9. Check whether the hydraulic transmission device is in good condition; 10. Check whether the main motor, main reducer, pinion and large gear ring of the main transmission system are normal and intact; whether the auxiliary motor and reducer of the auxiliary transmission are normal and intact; 11. Check whether the tooth ring and lubrication are good; 12. Check whether the pressing roller is normal; 13. Check whether the refractory materials in the kiln have fallen off and whether the kiln skin is tied.

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		1. 检查各处螺栓是否拧紧； 2. 检查回转窑润滑系统(4304-LUS-001)是否完好，各润滑点润滑脂是否加足； 3. 检查回转窑润滑系统(4304-LUS-001)冷却水是否正常； 4. 检查窑头密封不准有卡滞处； 5. 检查托轮调整是否正确，托轮与轮带表面是否均匀接触等； 6. 将系统设备仪表送电备妥，中控确认DCS系统正常； 7. 检查热电偶正常，其保护套突出窑衬表面，热电偶与套管上沿平齐； 8. 检查回转窑各处压力仪表装置是否正常，信号传输是否正常； 9. 检查液压挡轮装置是否完好； 10. 检查主传动系统的主电机、主减速器、小齿轮、大齿圈是否正常完好； 辅助传动的辅助电机、减速机是否正常完好； 11. 检查齿圈及润滑是否良好； 12. 检查压辊是否正常； 13. 检查窑内的耐火材料有没有脱落，是否结窑皮。
3	Inspect Kiln Head Cooling Fan (4304-BLO-005/006), Combustion Fan (4304-BLO-001/002) 检查窑头冷却风机 (4304-BLO-005/006)，燃烧风机 (4304-BLO-001/002)	1. Manual turning 2~3 turn, flexible rotation, no scraping, stuck and other abnormal phenomena; 2. The fan air outlet valve is normal, and the fan inlet valve is closed; 3. Whether the coupling concentricity meets the requirements; 4. Check whether the lubricating oil level of the fan bearing box is normal; 5. Close the power switch, first click (press the operation button and quickly press the stop button) to confirm that the steering is correct, no noise, no vibration and other abnormal conditions; 6. After the fan is started and running, observe the change of the fan sound, check the temperature and noise of the fan bearing and the motor bearing, and check the fan vibration (including the impeller). 1. 手动盘车2～3转，转动灵活，不得有刮蹭、卡住等异常现象； 2. 风机出风阀门正常，风机进口阀门关闭； 3. 联轴器同心度是否符合要求； 4. 检查风机轴承箱润滑油位是否正常； 5. 合上电源开关，先点动(按运转按钮后快速按停止按钮)确认转向正确、无杂音、无振动等异常情况；

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		6. 风机启动运行后，观察风机声音的变化，检查风机轴承、电机轴承的温度和噪声，检查风机振动(包括叶轮)。
4	Inspect Kiln Head Cover(4304-KHC-001) 检查窑头罩(4304-KHC-001)	1. Check the intact observation door (cover) of the kiln head; 2. Check that the kiln head sealing device is well sealed, and the sealing device is in good condition without falling off; 3. Whether the cooling air device is in good condition, and the bolts are not loose; 4. Check whether the refractory material of the kiln head falls off; 5. Check whether the fish scales of the kiln head cover are close to the rotary kiln cylinder, and whether the fish scale sealing device is adjustable and normal; 6. Whether the fish scale fastening wire rope is effective; 7. All kinds of instruments of the kiln head cover are normal and effective; 8. Check whether the inner screen installation is suitable; whether the sieve is worn, and whether the cooling water in the sieve is normal. 1. 检查窑头观察门(盖)完好; 2. 检查窑头密封装置密封良好，密封装置完好无脱落; 3. 冷却风装置是否保持良好状态，各处螺栓无松动; 4. 检查窑头耐火材料是否有脱落; 5. 检查窑头罩鱼鳞片是否紧贴回转窑筒体，鱼鳞片密封装置是否可调、是否正常; 6. 鱼鳞片紧固钢丝绳是否有效; 7. 窑头罩各种仪表正常有效; 8. 检查窑头罩内条格筛安装是否合适；条格筛有无磨损情况，条格筛内的冷却水是否正常。
5	Inspect Gas Bumer(4304-BUR-001) 检查煤气燃烧器(4304-BUR-001)	1. Check whether the burner (4304-BUR-001) of the kiln head is normal; 2. Check whether the fastening of the equipment is loose, and whether the forward extension and backward guide rail is normal; 3. Check whether the refractory material of the burner falls off; 4. Check whether all the wiring of the control box is correct; 5. Start the combustion fan first to determine the normal operation of the fan (if necessary, detect the air pressure of the burner air inlet, and the wind pressure is between 2 and 3 kPa); 6. Check whether the electronic gun lighter is normal; 7. Adjust the gas inlet pressure at about 10 kPa to detect whether the gas pipeline and valve group leak air; 8. Burner ignition steps a. Close the power supply switch;

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		b. Observe the panel indicator shows, that is, the power indicator is on and the stop indicator is on. If the fault indicator is on, press the fault reset button to remove the fault; c. Adjust the burner inlet air gate valve to 50% opening, and the gas inlet ball valve opening adjustment to about 30%; d. Start the blower in normal operation; e. Given the allowable startup conditions for remote transmission; f. For the first ignition, close the gas inlet ball valve (***), press the "gas turbine start" button, check whether the ignition procedure is correct; see the ignition from the rear of the furnace; g. The ignition procedure is correct, the ignition is normal, open the gas air inlet ball valve (***) to about 30% press the "gas start" button, the system enters the program automatic control, the ignition transformer ignition, ignition solenoid valve (***) open; if the program time, the fire is not lit, "fault indication" light, press "gas stop", "fault reset" button, after 30 seconds ignition; h. If the ignition is not done for many times (continuous ignition cannot exceed 5 times), the ignition should be stopped to avoid too much furnace intake and accidents; turn down the gas intake ball valve (***) or adjust the air intake, and then ignite (the gas valve (***) and air valve (***) during the system debugging, and the gas valve (****) opening is not allowed to adjust); i. The ignition program cycle is about 6 seconds, the operation indicator light is on, and only when the operation indicator is on, the ignition is successful; j. After the ignition operation is normal, increase the combustion load only adjust the air gate valve (***) opening, the air pressure change will adjust the opening of the air / combustion ratio valve, can realize the automatic ratio of air and gas; k. In normal operation, the compressed air of the musket must be kept unblocked, and do not allow to close the valve or lack of air operation, so as not to burn out the musket. 1. 检查窑头的燃烧器(4304-BUR-001)烧嘴是否正常; 2. 检查设备的紧固是否松动，前伸后退导轨是否正常; 3. 检查燃烧器耐火材料是否有脱落; 4. 检查控制箱所有接线是否正确; 5. 先启动燃烧风机，确定风机正常运转（必要时检测烧嘴空气进口风压，风压在2 ~ 3kPa之间）;

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		6. 检查电子点火枪是否正常； 7. 将燃气进口压力调节在10kPa左右，检测燃气管道、阀组是否漏气； 8. 燃烧器点火步骤 a. 合上电源开关； b. 观察面板指示灯显示，即电源指示灯亮，停炉指示灯亮，若故障指示灯亮，按下故障复位按钮，解除故障； c. 将燃嘴进口空气闸阀调到50%开度，燃气进口球阀开度调节为30%左右； d. 启动鼓风机正常运转； e. 给定远传允许启动条件； f. 第一次点火，先关闭燃气进气口小球阀(***)，按下“燃机启动”按钮，检查点火程序是否正确；从炉后部的观火孔可看见点火枪点火是否正常； g. 点火程序正确，点火枪点火正常，打开燃气进气口小球阀(***)到30%左右按下“燃机启动”按钮，系统进入程序自动检控，点火变压器工作点火，点火电磁阀(***)打开；若程序时间到，火没有点着，“故障指示”灯会亮，按下“燃机停止”、“故障复位”按钮，过30秒后再点火； h. 若多次点火不着(连续点火不能超过5次)，应停止点火以免炉窑进气太多，发生事故；调小燃气进气球阀(***)开度或调节空气进量，再进行点火(系统调试时已调好燃气阀(***)开度与空气闸阀(***)开度，燃气阀(***)开度不允许再调节)； i. 点火程序周期约6秒钟，运行指示灯亮，当且仅当运行指示亮后，点火才成功； j. 点火运行正常后，增加燃烧负荷只须调节空气闸阀(***)开度，空气导压管的压力变化会调节空/燃比例阀的开度，可以实现空气、燃气自动配比； k. 在正常运行中，点火枪的压缩空气一定要保持畅通，不允许关闭阀门或缺空气运行，以免烧坏点火枪。 l. 确认燃烧器(4304-BUR-001)前的煤气、闸阀(***)灵活好用，且全处在关闭状态，配套风机调节阀(***)、蝶阀(***)都灵活好用，且全处在关闭状态； m. 确认回转窑前煤气支管上排水器(***)畅通，且满水溢流，排水阀门(***)处在开启状态。
6	Check the other Equipment and Facilities 检查其它设备设	1. Lubrication System(4304-LUS-001) a. Remove the tubing of the lubrication system, close both ends, first dip 20% hydrochloric acid for 30 minutes, then neutralize the residual acid in the pipeline with 3%

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
	施	lime water, then rinse with water, blow dry with compressed air, add the same lubricating oil of the system, restore the installation of the installation should be strict and reliable; b. Clean the oil tank of the lubrication system, remove the debris and moisture in each corner with the dough, and then close it tightly, and inject clean lubricating oil as required; c. Start the lubrication system to cycle the oil to the rotary kiln system. First, wrap a layer of 0.150mm copper wire mesh at the tubing outlet. Then check the filter screen for impurities for 4 hours after the oil circulation. If the oil quality is good, remove the filter screen and restore the tubing system; d. After the lubrication system is opened, check whether there is abnormal vibration and abnormal sound, whether the oil leakage, whether the oil pressure is stable, whether the oil volume meets the requirements, whether the oil return is normal, whether the electrical system and temperature rise are normal. 1. 润滑系统(4304-LUS-001) a. 拆下润滑系统油管，封闭两端，先用20%盐酸浸渍30分钟后倒出，后用3%石灰水中和管道中的残酸，再用水冲洗，用压缩空气吹干后，加入系统相同规格型号的润滑油，恢复油管安装，安装后要严密可靠； b. 润滑系统油箱清洗干净，用面团沾除各角落的杂物与水分，然后封闭严密，按要求注入干净的润滑油； c. 启动润滑系统对回转窑系统进行油循环，先在油管出口包扎一层0.150mm的铜丝网，油循环进行4小时后检查滤网上是否有杂质，如油质良好，拆除滤网，恢复油管系统； d. 润滑系统开启后，检查是否有异常振动和异响，是否漏油，油压是否稳定，油量是否符合要求，回油是否正常，电气系统和温升是否正常。 2. Rap Device(4303-RAP-001) a. Check whether the connection of each part is fastened; b. Check whether the rotation direction is consistent with the mark direction. 2. 窑头溜槽(4304-CHE-002) a. 检查下料溜槽是否完好； b. 检查下料溜槽的堵料检测开关是否正常。

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		3. Bar screen a. Check whether the inner strip lattice sieve water beam device in the kiln head cover is working normally. 3. 条格筛 a. 检查窑头罩内条格筛水梁装置是否工作正常。 4. Valve a. Check whether the kiln head cooling fan (4304-BLO-005/006) (500BF) is normal; b. Check whether the combustion fan (4304-BLO-001/002) (500BF) is normal; c. Check whether the lubrication system (4304-LUS-001) cooling water inlet gate valve (65GA) is normal; d. Check whether the kiln tail release valve (***) is normal; e. Check whether the gas inlet ball valve (***), ignition solenoid valve (***), gas valve (****), air gate valve (***), glasses valve (***), primary butterfly valve (***) of the gas burner (4304-BUR-001) are normal. 4. 阀门 a. 检查窑头冷却风机(4304-BLO-005/006)进出口蝶阀(500BF)是否正常; b. 检查燃烧风机(4304-BLO-001/002)进出口蝶阀(500BF)是否正常; c. 检查润滑系统(4304-LUS-001)冷却水进口闸阀(65GA)是否正常; d. 检查窑尾放散阀(***)是否正常; e. 检查煤气燃烧器(4304-BUR-001)的燃气进气口小球阀(***)、点火电磁阀(***)、燃气阀(***)、空气闸阀(***)、眼镜阀(***)、送风一次蝶阀(***)等是否正常。
7	Inspect Materials Level 检查物料料位	1. Check whether the material is stored in the rotary kiln (4304-KLN-001); 2. Check whether the cooling circulating water pressure of the sieve water beam (4304-KHC-001) in the kiln hood is normal, and whether there is blocking material or foreign body on the sieve; 3. Check whether the gas and air pressure of the gas burner (4304-BUR-001) is normal, and whether there is any foreign body blockage in the gas nozzle; 4. Check whether the lubricating oil of the lubrication system (4304-LUS-001) is sufficient and whether the lubricating oil reflux condition is normal.

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		1. 检查回转窑(4304-KLN-001)内是否存料； 2. 检查窑头罩内(4304-KHC-001)条格筛水梁冷却循环水压力是否正常，条格筛上是否有堵料或异物； 3. 检查煤气燃烧器(4304-BUR-001)的煤气、空气压力是否正常，燃嘴是否有异物堵塞； 4. 检查润滑系统(4304-LUS-001)的润滑油是否充足，润滑油回流状况是否正常。
8	Start-Up Of the Kiln Head Post 窑头岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from back to front that is the reverse drive, Each equipment needs to be switched on later; At the same time, Confirm that the process behind the kiln head post has been started and has the feeding conditions: The cooling tower of flue gas treatment process (4306-TOW-001), reverse spray washer (4306-TOW-002), packing washing tower (4306-TOW-003), intercooler (4306-COO-005 / 006), intercooler (4306-COO-001 / 002 / 003 / 004), electric mist remover (4306-ESP-001 / 002) (except heating box), desulfurization tower (4306-TOW-004) are all opened, And open the corresponding spray circulation pump, Open one, 40Hz operation in the fan (4306-FAN-001 / 002 / 003), Ensure the negative pressure of rotary kiln tail-50~ -100Pa⟹ Open the cooling water of the lubrication system (4304-LUS-001) and start the lubrication system (4304-LUS-001)⟹ Open the combustion fan (4304-BLO-001 / 002)⟹15 minutes later, open the tail valve and open the side door of the kiln cover⟹ Diversion gas fills the main pipeline and the pipeline reaches a certain gas pressure; rear gas burner (4304-BUR-001) ignition⟹ After successful ignition, close the release valve and close the side door of the kiln hood⟹Start the kiln head cooling fan (4304-BLO-005 / 006) to operate at a low frequency⟹Open the cooling circulating water valve (4304-KHC 4-001) (150BF)⟹ Open the kiln tail cooling fan (4304-BLO-003 / 004)⟹ Open the main motor of the rotary kiln and run at 22Hz. 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，各设备都需要延时开启；同时，确认窑头岗位后面的工序已经开车并具备给料条件： 烟气处理工序的冷却塔(4306-TOW-001)、逆喷洗涤器(4306-TOW-002)、填料洗涤塔(4306-TOW-003)、间冷器

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		(4306-COO-005/006)、间冷器(4306-COO-001/002/003/004)、电除雾器 (4306-ESP-001/002)(除加热箱外)、脱硫塔(4306-TOW-004)全部逐次开启，并开启相应的喷淋循环泵，开启风机(4306-FAN-001/002/003)中的一台、40Hz运行，保证回转窑窑尾负压-50～-100Pa ⟹ 开启润滑系统(4304-LUS-001)冷却水，启动润滑系统(4304-LUS-001) ⟹ 开启燃烧风机(4304-BLO-001/002) ⟹ 十五分钟后，打开窑尾放散阀，打开窑头罩侧门 ⟹ 引煤气使煤气充满主管道，使管路达到一定的煤气压力；后煤气燃烧器(4304-BUR-001)点火 ⟹ 点火成功后关闭放散阀，关闭窑头罩侧门 ⟹ 启动窑头冷却风机(4304-BLO-005/006)低频率运行 ⟹ 开启窑头罩内(4304-KHC-001)条格筛水梁冷却循环水阀门(150BF) ⟹ 开启窑尾冷却风机(4304-BLO-003/004) ⟹ 开启回转窑主电机，22Hz运行。
9	The Kiln Head Post Stops 窑头岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order , Each equipment needs to stop: After the emptying of the kiln internal material, turn off the gas burner (4304-BUR-001) and stop the blower (4304-BLO-001 / 002) ⟹ At the early stage of the kiln shutdown, the kiln body (4304-KLN-001) is in a hot state, and the kiln body must be kept rotating ⟹ The cooling fan of the kiln head and the kiln tail sealing mechanism (4304-BLO-005 / 006 / 003 / 004) does not stop ⟹ When the temperature of the rotary kiln (4304-KLN-001) drops to about 200℃, stop the main motor of the rotary kiln (4304-KLN-001) ⟹ Stop the kiln tail cooling fan (4304-BLO-003 / 004) ⟹ Close the valve (4304-KHC 00-001) (150BF) ⟹ Stop kiln head cooling fan (4304-BLO-005 / 006) ⟹ Stop combustion fan (4304-BLO-001 / 002) ⟹ Close the lubrication system (4304-LUS-001) and the cooling water of the lubrication system (4304-LUS-001) ⟹ The cooling tower (4306-TOW-001), reverse spray washer (4306-TOW-002), packing washing tower (4306-TOW-003), intercooler (4306-COO-005 / 006), intercooler (4306-COO-001 / 002 / 003 / 003 / 004), electric mist remover (4306-ESP-001 / 002),

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		desulfurization tower (4306-TOW-004) are all stopped successively, And stop the corresponding spray circulation pump, Finally stop the fan (4306-FAN-001 / 002 / 003)⟹Stop carrier and gear station (4304-LUS-001)⟹Close the lubrication system (4304-LUS-001)⟹Close the cooling water of the lubrication system (4304-LUS-001) and close the lattice sieve (4304-KHC-001) in the kiln hood⟹Water beam cooling circulating water⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序顺序停车，各设备都需要延时停止： 窑内料排空后，关闭煤气燃烧器(4304-BUR-001)，停鼓风机(4304-BLO-001/002)⟹停窑初期，窑体(4304-KLN-001)处于炽热状态，须不停地转动窑体⟹窑头、窑尾密封机构冷却风机(4304-BLO-005/006/003/004)不停机⟹待回转窑(4304-KLN-001)温度降至200℃左右时，停止停止回转窑(4304-KLN-001)主电机⟹停止窑尾冷却风机(4304-BLO-003/004)⟹关闭窑头罩内(4304-KHC-001)条格筛水梁冷却循环水阀门(150BF)⟹停止窑头冷却风机(4304-BLO-005/006)⟹停止燃烧风机(4304-BLO-001/002⟹关闭润滑系统(4304-LUS-001)，关闭润滑系统(4304-LUS-001)冷却水⟹烟气处理工序的冷却塔(4306-TOW-001)、逆喷洗涤器(4306-TOW-002)、填料洗涤塔(4306-TOW-003)、间冷器(4306-COO-005/006)、间冷器(4306-COO-001/002/003/004)、电除雾器 (4306-ESP-001/002)、脱硫塔(4306-TOW-004)全部逐次停止，并停止相应的喷淋循环泵，最后停止风机(4306-FAN-001/002/003)⟹停止托轮和挡轮稀油站(4304-LUS-001)⟹关闭润滑系统(4304-LUS-001)⟹关闭润滑系统(4304-LUS-001)冷却水，关闭窑头罩内(4304-KHC-001)条格筛⟹水梁冷却循环水⟹打扫现场卫生。

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		2. Emergency Stop a. When an emergency occurs, Stop rotary kiln (4304-KLN-001), Stop kiln head cooling fan (4304-BLO-005/006), gas burner (4304-BUR-001), combustion fan (4304-BLO-001/002), Lubrication System (4304-LUS-001), fan (4304-BLO-005/006), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close the kiln head cooling fan (4304-BLO-005/006) (500BF), Combustion fan (4304-BLO-001/002) (500BF), Lubrication System (4304-LUS-001) cooling water inlet gate valve (65GA), Kiln tail release valve (***), Gas burner (4304-BUR-001) gas inlet ball valve (***), ignition solenoid valve (***), gas valve (***), air gate valve (***), glasses valve (***), air supply primary butterfly valve (***), at the same time, the upper level of equipment all need to stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, first heat up, when the smoke kiln tail degree reaches 900℃, start to feed, the slowest machine speed, the kiln tail temperature reaches normal 20 minutes, depending on the dry ball roasting, gradually restore the normal machine speed; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, First start the rotary kiln, and after the pellets in the kiln is discharged, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，停转回转窑(4304-KLN-001)，停止窑头冷却风机(4304-BLO-005/006)、煤气燃烧器(4304-BUR-001)、燃烧风机(4304-BLO-001/002)、润滑系统(4304-LUS-001)、风机(4304-BLO-005/006)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭窑头冷却风机(4304-BLO-005/006)进出口蝶阀(500BF)，燃烧风机(4304-BLO-001/002)进出口蝶阀(500BF)，润滑系统(4304-LUS-001)冷却水进口闸阀(65GA)，窑尾放散阀(***)，燃烧器(4304-BUR-001)的燃气进气口小球阀(***)、点火电磁阀(***)、燃气阀(***)、空气闸阀(***)、眼镜阀(***)、送风一次蝶阀(***)，同时，上一级设备均需停车； b. 事故处理后，设备恢复正常，如仍需要进行生产，

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

№	Stage	Key Points
		首先升温，待烟窑尾温度达到900℃以上，开始投料，机速最慢，窑尾温度达到正常20分钟后，视干球焙烧情况，逐步恢复正常机速； c. 事故处理后，设备恢复正常，如不需要进行生产，首先启动回转窑，待窑内铁球团排净后，按照正常停车顺序进行操作。
10	Permits & Isolation Sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

8. Kiln Head Post 窑头岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

9. Appendix

This publication may not be reproduced, stored in the system, transferred in any form by any means, electronic, mechanical, photocopying, recorded, or otherwise without the permission of PT. Merdeka Tsingshan Indonesia.

Standard Operating Procedure
标准操作程序

Kiln head posts in the chloride plant
CP 窑头岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

Document Owner 程序持有人	Song Yuhua 宋裕华	Date: 日期: 2024.4.24	Signature 签字	
Processing Superintendent 氯化高级副经理	Cao Guibin 曹桂宾	Date: 日期: 2024.4.24	Signature 签字	
Processing Senior Manager 氯化高级经理	Li Jianjun 李建军	Date: 日期: 2024.04.24	Signature 签字	
Safety Department Manager 安全部经理	Airlangga Putra R.J	Date: 日期: 24.04.24	Signature 签字	
Deputy General Manager 副总经理	Li Xinhua 李新华	Date: 日期 2024.4.24	Signature 签字	
General Manager 总经理	Clark Wu 吴迎林	Date: 日期: 2024.4.24	Signature 签字	